



Pranav Khare <khare.pranav@gmail.com>

Invitation to Review for IEEE Access Access-2024-49997

1 message

IEEE Access <onbehalf@manuscriptcentral.com>

Wed, Jan 29, 2025 at 1:41 AM

Reply-To: ieeeaccess2@ieee.org

To: khare.pranav@gmail.com

29-Jan-2025

Dear Mr. Khare:

The manuscript entitled "Leveraging Cross-Project Similarity for Data Augmentation and Security Bug Report Prediction" has been submitted to IEEE Access. Since your expertise is relevant to this article, we would be grateful if you would consider reviewing it.

The abstract appears at the end of this letter, along with the names of the authors. Please let me know within 2 days if you will be able to accept my invitation to review. If you agree to review, we ask that you provide your review comments within **7 days** to assist IEEE Access in maintaining its expedited peer review process of 4 to 6 weeks. If you are interested in reviewing but need additional time to complete the review, we can grant a reasonable deadline extension upon request. For additional information on IEEE Access and tips on completing a quality review, please visit our [Reviewer Best Practices guide](#).

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Thank you for your assistance with the peer review of this IEEE Access article.

Sincerely,

Prof. Mehrdad Saif
Editor-in-Chief, IEEE Access

ieeeaccess2@ieee.org

MANUSCRIPT DETAILS

TYPE: Research Article

TITLE: Leveraging Cross-Project Similarity for Data Augmentation and Security Bug Report Prediction

AUTHORS: Ji, Jinfeng; Yang, Geunseok

ABSTRACT: Bug reports are generally categorized into two types: security and non-security. However, not all bug reporters possess the expertise needed for accurate classification. As a result, some security-related bug reports are often mistakenly labeled as non-security. Identifying these mislabeled security bug reports within the non-security category places an additional burden on the relevant staff. Security bugs are particularly critical because they can disrupt the normal operation of systems and compromise user data security. To address this issue, we propose a method for predicting security bug reports. Bug reports were extracted from four open-source projects: Ambari, Camel, Derby, and Wicket. The number of security bug reports for the four projects is 56, 74, 179, and 47, and the number of non-security bug reports is 944, 926, 821, and 953, respectively, they are an imbalanced dataset. So we use the similarity across projects to perform data augmentation after preprocessing the data. By combining security bug reports with their similar counterparts across projects, our method achieved optimal performance, demonstrating the model's effectiveness. The F1 score of this method ranges from 0.60 to 0.98, with LSTM on Ambari and GRU on both Ambari and Camel achieving the highest performance of 0.98. The average F1 score of the method is 0.77, representing a significant improvement in prediction accuracy.

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